XINGE GUO

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RESEARCH INTERESTS

Computer vision and machine learning for medical image analysis, with a focus on **Representation Learning**, **Image Segmentation**, **Clinical Decision Making** and **Treatment Planning**.

EDUCATION

Duke University, Durham, NC, USA **Aug 2025 – Present**

M.S. in Biomedical Engineering · Advisor: Prof. Sina Farsiu (VIP Lab) *GPA: 3.81 / 4.00*

- **Fellowships:** Research Fellowship for BME MS/MEng Students (Fall 2025 & Spring 2026); Merit-based Scholarship (Fall 2025)

Shenzhen University, Shenzhen, China **Sep 2021 – Jun 2025**

B.Eng. in Biomedical Engineering, Class of Excellence *GPA: 3.91 / 4.50*

- **Honors:** Top Innovative Talent Scholarship (Dec 2022, Dec 2023, Dec 2024); Full Scholarship for Class of Excellence (Sep 2021)

PUBLICATIONS

1. Xiao F, Hu P, Xu L, **Guo X**, Qin G, Shen Y, Fang C, Zhang R, He C, Farsiu S. "*Beyond Ground-Truth: Leveraging Image Quality Priors for Real-World Image Restoration*." **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.**
2. **Guo X**, Liu W, Cui Z. "*Efficient Semi-supervised Tooth Instance Segmentation in Panoramic X-Rays Using ResUnet50 and SAM Networks*." **International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2024,** pp. 131–145. *(First Author)*
3. Dai J, **Guo X**, Zhang H, et al. "*Cone-beam CT Landmark Detection for Measuring Basal Bone Width: A Retrospective Validation Study*." **BMC Oral Health, 2024,** 24(1): 1091. *(Co-First Author)*

RESEARCH EXPERIENCE

Vision and Image Processing (VIP) Laboratory, Duke University **Sep 2025 – Present**

Graduate Researcher · Advisor: Prof. Sina Farsiu *Durham, NC, USA*

Project 1 — Leveraging Image Quality Priors for Real-World Image Restoration (CVPR 2026)

- Co-developed IQPIR, a framework that replaces pixel-level ground-truth supervision with a **learned image quality prior** to guide real-world image restoration, addressing the systematic bias of ground-truth-based metrics in optimizing restoration models.
- Contributed to method design, experiments on multiple restoration benchmarks, and ablation studies analyzing how the quality prior shapes the restoration trajectory.

Project 2 — Segmentation Difficulty Benchmark via VLM (Ongoing)

- Building a benchmark for **predicting segmentation difficulty from input images alone**, reframing segmentation difficulty as a task-relative image quality prior.
- Designing a VLM-based agent that interacts with multiple segmentation models to estimate per-image difficulty and identify likely failure regions before deployment.

Project 3 — Hyperspherical Feature Representations for Segmentation (Ongoing)

- Exploring **hyperspherical feature representations** as a geometric prior on the embedding space to improve segmentation accuracy and robustness, motivated by the limitations of unconstrained Euclidean feature spaces in medical imaging.

Department of Orthopaedic Surgery, Duke University School of Medicine **Dec 2025 – Present**

Research Collaborator · Advisor: Dr. Brian C. Lau, MD — *Durham, NC, USA*

Project — Deep Learning-Based Glenoid and Humerus Bone Loss Measurement and Severity Stratification on 3D CT in Shoulder Instability (Ongoing)

- Developing 3D CT segmentation and quantitative measurement pipelines for glenoid and humeral bone loss assessment, in close collaboration with orthopaedic surgeons. Manuscript in preparation.

Medical Ultrasound Engineering Lab (National-Regional Key), Shenzhen University — **Mar 2024 – Apr 2025**

Undergraduate Researcher · Advisor: Prof. Chien Ting Chin — *Shenzhen, China*

Project — Coordinate Classification Deep Learning Network for Left Ventricle Landmark Detection (Ongoing, manuscript in preparation)

- Developed a lightweight, end-to-end deep learning network reformulating LV landmark detection on cardiac ultrasound as a coordinate classification problem, improving precision and inference efficiency over heatmap regression baselines.

Department of Radiology and Nuclear Medicine, Erasmus University Medical Center — **Jul 2024 – Sep 2024**

Research Intern · Mentor: Dr. Xianjing Liu — *Rotterdam, the Netherlands*

Project — GWAS on AI-Derived Phenotypes from Structural Brain MRI in UK Biobank

- Implemented a 3D convolutional autoencoder for representation learning on ~40,000 structural brain MRI scans from the UK Biobank, supporting downstream genome-wide association analyses on AI-derived imaging phenotypes.

Medical AI Laboratory, School of Biomedical Engineering, Shenzhen University — **Aug 2022 – Oct 2024**

Undergraduate Researcher · Advisor: Prof. Bingsheng Huang — *Shenzhen, China*

Project 1 — Semi-Supervised Tooth Instance Segmentation on Panoramic X-Rays (MICCAI 2024, First Author)

- Designed a semi-supervised framework combining **ResUnet50** with the **Segment Anything Model (SAM)** to improve tooth instance segmentation in 2D panoramic X-rays under limited annotation, achieving competitive performance against fully supervised baselines.

Project 2 — CBCT Landmark Detection for Basal Bone Width Measurement (BMC Oral Health 2024, Co-First Author)

- Developed a U-Net-based deep learning model for precise orthodontic landmark detection on CBCT images, in collaboration with orthodontic surgeons; retrospectively validated on clinical cases for basal bone width assessment.

SELECTED COURSEWORK

- **Duke (Graduate):** AI Image Assessment (Independent Study with Prof. Farsiu, A+), Signal Processing and Applied Math (A), Ultrasound Imaging (A), Digital Image Processing (Fall 2026, in progress)
- **Shenzhen U (Undergraduate):** Principles of Medical Imaging (A+), Medical Ultrasound Technology (A), Medical Signal Processing (A), Signals and Systems (A), Digital Electronics (A+), Probability and Statistics (A), Linear Algebra (A+), Calculus (A+), Python Programming (A)

TECHNICAL SKILLS

- **Programming:** Python (primary), MATLAB, C/C++, VHDL, Assembly
- **Languages:** English (Fluent, IELTS 7.5), Mandarin (Native), Cantonese (Native)

OTHER

- **Piano Performance:** Associate Diploma (ALCM), London College of Music, University of West London (Apr 2022).